1.5 Solving Proportions

Name _____ Period ____ Date _____

Algebra I 1.5 · Student sheet

I can…

- solve proportions.
- convert units and rates.
- compare rates to solve real-life problems.

Vocabulary · fill in the missing word

Ratio — a _____ of two quantities by division

Proportion — an _____ stating that two ratios are equal

Cross Products Property — if $\frac{a}{b}$ = _____, then ad = bc

Unit rate — a rate with a _____ of 1 unit

Conversion factor — a ratio equal to 1, such as 12 in./1 ft, used to _____ units

Part 1 · Best buy

Product	Option 1	Unit price	Option 2	Unit price	Better buy
Cereal	12 oz for $3.60		18 oz for $4.86		
Chips	10 oz for $3.50		16 oz for $5.28		
Juice	64 fl oz for $4.48		32 fl oz for $2.08		

How did you decide? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Proportion: A proportion states that two _____ are equal.

Cross Products Property: Cross products: if a/b = c/d, then _____ = _____.

Example 1

Solve $\frac{x}{12} = \frac{5}{8}$.

Example 2

A 6-foot person casts a 4-foot shadow. At the same time, a tree casts a 22-foot shadow. How tall is the tree?

Converting units: To convert units, multiply by a _____ factor equal to 1.

Example 3

a. Convert 2.5 kilometers to meters.

b. Convert 60 miles per hour to feet per second.

Example 4

Runner A runs 400 meters in 50 seconds. Runner B runs 1 mile (about 1609 meters) in 4 minutes. Who is faster?

Part 3 · Partner practice

Partner 1

Solve $\frac{x+2}{9} = \frac{4}{6}$.

Partner 2

Convert 45 feet per minute to inches per second.